Chen Xie

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Research Position

- **Assistant Research Scientist** — Baltimore, MD
  The Johns Hopkins University — *2023 - present*
- Graduate Student Researcher — Marseille, France
  Aix-Marseille Université — *2020 - 2023*
- Summer Student internship — Dwingeloo, The Netherlands
  ASTRON/JIVE — *2017*

Education

- **Aix-Marseille Université** — Marseille, France
  ***PhD** in Astronomy* — *awarded in Feb 2024*
  - PhD Thesis: Development of advanced post-processing methods for the direct detection of exoplanets with ground-based high-contrast imagers
  - Thesis Advisor: Dr. Arthur Vigan, Dr. Elodie Choquet
- **Leiden University** — Leiden, The Netherlands
  ***M.Sc.** in Astronomy* — *Feb 2020*
  - Major Thesis: VLT/MUSE as a high-contrast imager
  - Thesis Advisors: Prof. Matthew Kenworthy, Dr. Jos de Boer, and Dr. Sebastiaan Haffert
  - Minor Thesis: Radio Observations of Frontier Fields Clusters Abell S1063 and Abell 370: The Discovery of New Radio Halos
  - Thesis Advisor: Prof. Reinout van Weeren
- **Xiamen University** — Xiamen, China
  ***B.Sc.** in Physics* — *June 2016*
  - Bachelor Thesis: On the Host Galaxy of GRB 150101B
  - Thesis Advisor: Prof. Taotao Fang

Selected Grants & Awards

Total awards for JWST programs as Grant Admin PI: **$149,369** . . . . Cycle 3 – 4
Poster Prize, Lyot conference 2022
Oort Scholarship (**€22,000 + tuition waiver**), Leiden University . . . . 2018 – 2020
First Prize of Lin Qiao Prize for Excellent Undergraduate Research Project, Peking University 2016

Mentoring

5. Chen, J.-Y. (MSc student, Xiamen Univ., ongoing) – SAFFORN survey, spiral motion measurement
4. Zhong, H.-S. (PhD student, Sun Yat-sen Univ., ongoing) – Reflectance spectra of HD 32297 debris disk; paper in prep.
3. Xie, C.-Y. (PhD student, Univ. of Arizona, 2024) – publication in Universe, 10, 465 (2024).
2. Ahmad, Y. (MSc student, EuroPhotonics, 2022–23) – developing image-processing techniques
1. Rahman, S. (MSc student, EuroPhotonics, 2021–22) – developing image-processing techniques

Publications (135 citations from 9 first-/corresponding-author papers)

Total 20 papers, incl. 1 in Nature (1st authored), 1 in Science (3rd authored, in preview), 1 in Nature Astronomy

1st-/Corresponding-authored:

1. **Xie, C.**, et al. *Water ice in the debris disk around HD 181327*, **Nature**, 641, 608 (2025).
   Research Highlight: water ice discovery
2. **Xie, C.**, et al. *Reference-star differential imaging on SPHERE/IRDIS*, A&A, 666, A32 (2022).
   Research Highlight: new archival RDI with 2.1× improvement
3. **Xie, C.**, et al. *Searching for proto-planets with MUSE*, A&A, 644, A149 (2020).
   Research Highlight: first MUSE survey
4. **Xie, C.**, et al. *DESTINYS: Dynamical evidence of a spiral-arm-driving protoplanet and Gap-Opening Protoplanet from SAO 206462 Spiral Motion*, Universe, 10, 465 (2024).
5. Ren, B.B., **Xie, C.**, et al. *A companion in V1247 Ori supported by spiral motion*, A&A, 681, L2 (2024).
6. **Xie, C.**, et al. *Dynamical detection of a companion driving a spiral arm*, A&A, 675, L1 (2023).
7. **Xie, C.**, et al. *MUSE view of asymmetric jet from HD 163296*, A&A, 650, L6 (2021).
8. **Xie, C.**, et al. *The discovery of radio halos in the Frontier Fields clusters Abell S1063 and Abell 370*, A&A, 636, A3 (2020)
9. **Xie, C.**, et al. *On the Host Galaxy of GRB 150101B and the Associated Active Galactic Nucleus*, ApJL, 824, L17 (2016)

2nd/3rd authored:

10. Betti, S.K., Chen, C.H., **Xie, C.**, et al. *Detection of ices in the debris disk around β Pictoris*, **Science** (under review since Feb. 2025).
11. Matthews, E.C., Bonnefoy, M., **Xie, C.**, et al. *First scattered-light images of HD 112810 debris disk*, A&A, 679, A58 (2023).
12. Uyama, T., **Xie, C.**, et al. *Keck/OSIRIS Paβ constraints on PDS 70b*, AJ, 162, 214 (2021).

Contributing authored:

13. Worthen, K., Chen, C.H., Brittain, S.D., Najita, J.R., **Xie, C.** et al. *Fluorescently excited CO emission in the 49 Ceti debris disk spatially resolved by JWST/NIRSpec*, **Nature Astronomy**, (2025)
14. Zhong, H.-S., Ren, B., Ma, B., **Xie, C.** et al. *Multiband reflectance and shadowing of the protoplanetary disk RX J1604.3-2130 in scattered light*, A&A, 684, A168 (2024)
15. Lu, C.X., et al. (incl. **Xie, C.**) *JWST/NIRSpec Detects Warm CO Emission in the Terrestrial-Planet Zone of HD 131488*, ApJ, (under review since June. 2025).
16. Engler, N. et al. (incl. **Xie, C.**) *Characterization of debris disks observed with SPHERE*, A&A, (under review since April. 2025).

White papers (non peer-reviewed):

17. **Xie, C.** et al. *The transmission spectrum of a terrestrial planet-forming zone*, Roman Coronagraph White Papers (2025).

18. Chen, C.H., et al. (incl. **Xie, C.**). *Water Delivery in the η Crv Habitable Zone*, Roman Coronagraph White Papers (2025).
19. Lewis, B.L., et al. (incl. **Xie, C.**). *Characterizing Known Debris Disk HR 4796A in Polarized Scattered Light*, Roman Coronagraph White Papers (2025).
20. Hom, J., et al. (incl. **Xie, C.**). *A First Look at Bright Debris Disks with HLC Band 1*, Roman Coronagraph White Papers (2025).

Leadership

Spiral motion lead, VLT/SPHERE SAFFRON survey (PI: B. Ren; 69.5 hrs; 40+ members), primary goal

Selected Meetings since 2022

Gordon Research Conferences, Origins of Solar Systems, ... Poster, South Hadley, MA, 2025
National Capital Area Disks Meeting, ... **Talk**, Washington, DC, 2025
STScI JWST science conversation, ... **Invited talk**, Baltimore, MD, 2025
Earth & Planets Laboratory Seminar, ... **Invited talk**, Washington, DC, 2025
STScI-JHU ExoJamboree, ... **Talk**, Baltimore, MD, 2024
Dust Devils workshop, ... **Talk**, Tucson, AZ, 2024
AO4ELT, ... Poster, Marseille, 2023
Observatory of the Côte d'Azur Seminar, ... **Invited talk**, Nice, 2023
ETH Zürich Exoplanets & Habitability Seminar, ... **Invited talk**, Zürich, 2022
Observatory of the Côte d'Azur Seminar, ... **Invited talk**, Nice, 2022
Debris discs: At Home and Abroad, ... **Talk**, Jena, 2022
Spirit of Lyot 2022, ... Poster, Leiden, 2022

Service

Member, Roman Coronagraph Community Participation Program .... 2025 –
Referee, AAS journals .... 2024 –
Referee, A&A .... 2023 –

Press & Media

NASA | Another First: NASA Webb Identifies Frozen Water in Young Star System 2025
BBC | This baby star is hiding water ice, and maybe the seeds of a new earth 2025
Space | James Webb Space Telescope discovers an alien planetary system's icy edge ... 2025
APOD | HD 163296: Jet from a Star in Formation 2021

Successful Telescope Proposals

Total 37.7 hrs in JWST, 4 orbits in HST, 68.5 hrs in VLT, 23.6 hrs in radio (VLA, GMRT, LOFAR)
Incl. 8.9 hrs in JWST (US admin PI), 2 hrs in VLTI (PI), 3 hrs in MUSE (d-PI), 15.6 hrs in radio (PI)

- HST (GO 18026): Testing volatile transportation in the epoch of terrestrial planet formation
  Col-I, 4 orbits *2025*
- JWST (GO 7313): Determining the Origin of the Gas in the 49 Ceti Debris Disk
  Col-I, 7.2 hrs *2025*
- JWST (GO 6991): Testing planet-disk interaction theory
  Admin-PI, 8.9 hrs, $106,069 *2025*

- JWST (GO 6940): Determining the Origin of Water Ice in the β Pictoris Disk
  Col-I, 13.7 hrs *2025*
- VLTI/GRAVITY: Pinning down the orbit of HD 100453 B
  PI, 1 hr *2025*
- VLT/SPHERE: Locating hidden giant planets
  Co-I, 5 hrs *2025*
- JWST (GO 5261): Confirming the youngest gap-opening protoplanet
  Col-I, 7.9 hrs, $43,300 *2024*
- VLTI/GRAVITY: Pinning down the orbit of HD 100453 B
  PI, 1 hr *2024*
- VLTI/GRAVITY: Confirming the youngest gap-opening protoplanet
  Co-I, 3 hrs *2024*
- VLT/VISIR: Do exocomets release dust?
  Co-I, 1 hr *2023*
- VLT/SPHERE: Locating hidden giant planets: the SAFFRON survey
  Co-I, 54.5 hrs *2023*
- VLT/MUSE: Astrochemistry hints at the presence of a young accreting planet
  d-PI, 3 hrs *2022*
- Subaru: Astrochemistry hints at the presence of a young accreting planet
  Co-I, 0.5 night *2022*
- GMRT: Testing turbulent re-acceleration in Abell S1063
  PI, 10 hrs *2019*
- VLA: Testing turbulent re-acceleration in Abell S1063
  PI, 5.6 hrs *2019*
- LOFAR: Observing the previously undetected diffuse radio emission
  Col-I, 8 hrs *2017*
- CFHT: Studying the origin of X-ray arcs in M51b
  Col-I, 1 hr *2016*